Figure (d) shows the display optimized to view pixel values around digital number 75. This is done by turning up the *contrast*, resulting in the output transform increasing in *slope*. For example, the stored pixel values of 71 and 75 become 100 and 116 in the display, making the contrast a factor of four greater. Pixel values between 46 and 109 are displayed as the blackest black, to the whitest white. The price for this increased contrast is that pixel values 0 to 45 are saturated at black, and pixel values 110 to 255 are saturated at white. As shown in (d), the increased contrast allows the triangles in the left square to be seen, at the cost of saturating the middle and right squares.

Figure (e) shows the effect of increasing the contrast even further, resulting in only 16 of the possible 256 stored levels being displayed as nonsaturated. The brightness has also been decreased so that the 16 usable levels are centered on digital number 175. The details in the center square are now very visible; however, almost everything else in the image is saturated. For example, look at the noise around the border of the image. There are very few pixels with an intermediate gray shade; almost every pixel is either pure black or pure white. This technique of using high contrast to view only a few levels is sometimes called a **grayscale stretch**.

The contrast adjustment is a way of *zooming in* on a smaller range of pixel values. The brightness control *centers* the zoomed section on the pixel values of interest. Most digital imaging systems allow the brightness and contrast to be adjusted in just this manner, and often provide a graphical display of the output transform (as in Fig. 23-12). In comparison, the brightness and contrast controls on television and video monitors are *analog circuits*, and may operate differently. For example, the contrast control of a monitor may adjust the gain of the analog signal, while the brightness might add or subtract a DC offset. The moral is, don't be surprised if these analog controls don't respond in the way you think they should.

Grayscale Transforms

The last image, Fig. 23-12f, is different from the rest. Rather than having a slope in the curve over *one* range of input values, it has a slope in the curve over *two* ranges. This allows the display to simultaneously show the triangles in both the left and the right squares. Of course, this results in saturation of the pixel values that are *not* near these digital numbers. Notice that the slide bars for contrast and brightness are not shown in (f); this display is beyond what brightness and contrast adjustments can provide.

Taking this approach further results in a powerful technique for improving the appearance of images: the **grayscale transform**. The idea is to increase the contrast at pixel values of interest, at the expense of the pixel values we don't care about. This is done by defining the relative importance of each of the 0 to 255 possible pixel values. The more important the value, the greater its contrast is made in the displayed image. An example will show a systematic way of implementing this procedure.